We estimate the regression

$$y_i = X_i'\beta + u_{p(i)} + \varepsilon_i \tag{1}$$

where X_i are controls and the u_p are fixed effects for an individual's plan $p(i)$. Let $\hat{u}_p$ be our estimate of u_p, and write

$$\hat{u}_p = u_p + \nu_p \tag{2}$$

$$u_p = w_{c(p)} + \eta_p \tag{3}$$

where ν_p denotes estimation error and η_p is the residual plan effect within its county $c(p)$. Let N_p denote the number of people in plan p and $N_c = \sum_{p:c(p)=c} N_p$ be the number of people in county c. Suppose $w_c \sim \mathcal{N}(\mu, \sigma_w^2)$, $\eta_p \sim \mathcal{N}(0, \sigma_\eta^2)$, and $\nu_p \sim \mathcal{N}(0, \sigma_\varepsilon^2/N_p)$, all *iid* and mutually independent, and let

$$\begin{aligned}\hat{w}_c &= \frac{1}{N_c}\sum_{p:c(p)=c} N_p \hat{u}_p \\ &= w_c + \frac{1}{N_c}\sum_{p:c(p)=c} N_p(\eta_p + \nu_p)\end{aligned} \tag{4}$$

be the enrollment-weighted average plan effect estimate in county c. Then

$$\begin{bmatrix} u_p \\ \hat{u}_p \\ \hat{w}_{c(p)} \end{bmatrix} \sim N\left(\begin{bmatrix} \mu \\ \mu \\ \mu \end{bmatrix}, \begin{bmatrix} \sigma_w^2 + \sigma_\eta^2 & \sigma_w^2 + \sigma_\eta^2 & \sigma_w^2 + \sigma_\eta^2 \frac{N_p}{N_{c(p)}} \\ \sigma_w^2 + \sigma_\eta^2 & \sigma_w^2 + \sigma_\eta^2 + \sigma_\varepsilon^2/N_p & \sigma_w^2 + \sigma_\eta^2 \frac{N_p}{N_{c(p)}} + \sigma_\varepsilon^2/N_{c(p)} \\ \sigma_w^2 + \sigma_\eta^2 \frac{N_p}{N_{c(p)}} & \sigma_w^2 + \sigma_\eta^2 \frac{N_p}{N_{c(p)}} + \sigma_\varepsilon^2/N_{c(p)} & \sigma_w^2 + \sigma_\eta^2 M_{c(p)} + \sigma_\varepsilon^2/N_{c(p)} \end{bmatrix} \right)$$

where $M_c = \sum_{p:c(p)=c}(N_p/N_c)^2$ is county c's Herfindahl index. We then have

$$\begin{aligned} E[u_p \mid \hat{u}_p, \hat{w}_{c(p)}] &= \mu + \begin{bmatrix} \sigma_w^2 + \sigma_\eta^2 & \sigma_w^2 + \sigma_\eta^2 \frac{N_p}{N_{c(p)}} \end{bmatrix} \begin{bmatrix} \sigma_w^2 + \sigma_\eta^2 + \sigma_\varepsilon^2/N_p & \sigma_w^2 + \sigma_\eta^2 \frac{N_p}{N_{c(p)}} + \sigma_\varepsilon^2/N_{c(p)} \\ \sigma_w^2 + \sigma_\eta^2 \frac{N_p}{N_{c(p)}} + \sigma_\varepsilon^2/N_{c(p)} & \sigma_w^2 + \sigma_\eta^2 M_{c(p)} + \sigma_\varepsilon^2/N_{c(p)} \end{bmatrix}^{-1} \begin{bmatrix} \hat{u}_p - \mu \\ \hat{w}_{c(p)-\mu} \end{bmatrix} \\ &= \lambda_{1p}\hat{u}_p + \lambda_{2p}\hat{w}_{c(p)} + (1 - \lambda_{1p} - \lambda_{2p})\mu, \end{aligned} \tag{5}$$

where (omitting some tedious algebra)

$$\lambda_{1p} = \frac{\sigma_\eta^2 \left(\left(1 + M_{c(p)} - 2\frac{N_p}{N_{c(p)}}\right)\sigma_w^2 + \left(\sigma_\eta^2 M_{c(p)} + \sigma_\varepsilon^2/N_{c(p)}\right) - \frac{N_p}{N_{c(p)}}\left(\sigma_\eta^2 \frac{N_p}{N_{c(p)}} + \sigma_\varepsilon^2/N_{c(p)}\right)\right)}{\left(\sigma_w^2 + \sigma_\eta^2 + \sigma_\varepsilon^2/N_p\right)\left(\sigma_w^2 + \sigma_\eta^2 M_{c(p)} + \sigma_\varepsilon^2/N_{c(p)}\right) - \left(\sigma_w^2 + \sigma_\eta^2 \frac{N_p}{N_{c(p)}} + \sigma_\varepsilon^2/N_{c(p)}\right)^2} \tag{6}$$

$$\lambda_{2p} = \frac{\sigma_w^2 \sigma_\varepsilon^2 \frac{N_{c(p)} - N_p}{N_p N_{c(p)}}}{\left(\sigma_w^2 + \sigma_\eta^2 + \sigma_\varepsilon^2/N_p\right)\left(\sigma_w^2 + \sigma_\eta^2 M_{c(p)} + \sigma_\varepsilon^2/N_{c(p)}\right) - \left(\sigma_w^2 + \sigma_\eta^2 \frac{N_p}{N_{c(p)}} + \sigma_\varepsilon^2/N_{c(p)}\right)^2}. \tag{7}$$

Note that as $\sigma_w \to 0$ (no county effects), we have $\lambda_{2p} \to 0$ and

$$\begin{aligned}\lambda_{1p} &\to \frac{\sigma_\eta^2 \left(\sigma_\eta^2 M_{c(p)} + \sigma_\varepsilon^2/N_{c(p)}\right) - \frac{N_p}{N_{c(p)}}\left(\sigma_\eta^2 \frac{N_p}{N_{c(p)}} + \sigma_\varepsilon^2/N_{c(p)}\right)}{\left(\sigma_\eta^2 + \sigma_\varepsilon^2/N_p\right)\left(\sigma_\eta^2 M_{c(p)} + \sigma_\varepsilon^2/N_{c(p)}\right) - \left(\sigma_\eta^2 \frac{N_p}{N_{c(p)}} + \sigma_\varepsilon^2/N_{c(p)}\right)^2} \\ &= \frac{\sigma_\eta^2}{\sigma_\eta^2 + \sigma_\varepsilon^2/N_p},\end{aligned} \tag{8}$$

our conventional plan shrinkage coefficient. Similarly when $\sigma_\eta \to 0$ (only county effects), $\lambda_{1p} \to 0$ and

$$\begin{aligned}\lambda_{2p} &\to \frac{\sigma_w^2 \sigma_\varepsilon^2 \frac{N_{c(p)} - N_p}{N_p N_{c(p)}}}{\left(\sigma_w^2 + \sigma_\varepsilon^2/N_p\right)\left(\sigma_w^2 + \sigma_\varepsilon^2/N_{c(p)}\right) - \left(\sigma_w^2 + \sigma_\varepsilon^2/N_{c(p)}\right)^2} \\ &= \frac{\sigma_w^2}{\sigma_w^2 + \sigma_\varepsilon^2/N_{c(p)}}\end{aligned} \tag{9}$$

which is the conventional shrinkage coefficient we would apply to the county averages by themselves.